ZEJIA DUAN

zoeduan@uchicago.edu | github.com/zoe-duan

SKILLS & INTERESTS

- **Research Interests:** Public & Labor Economics, Corporate & Household Finance, Applied Microeconomics
- **Skills:** Research Design, Literature Reviews, Quantitative Methods (DID, Bayesian, Causal Inference, Regressions, IV), Data Collection & Statistical Analysis, Data Visualization, Machine Learning, Large Language Models, Academic & Professional Writing, Project Management, Teamwork & Collaboration
- **Technical:** Python, Stata, R, SQL; Data and Research Tools: iFinD, CSMAR, Office 365 (Excel), Git/GitHub
- **Languages:** English: IELTS Overall 7.0 (Listening 7.5, Reading 8.0, Writing 6.5, Speaking 6.5)

EDUCATION

University of Chicago | Chicago, IL

M.A. in the Social Sciences - Quantitative Methods and Social Analysis, Expected June 2027

- Current Coursework (Fall 2026): Statistical Theory & Methods, Marketing Strategy, Social Science Analysis

Renmin University of China (RUC) | Beijing, China

B.A. in Philosophy, Minor in Economics, June 2026

- Selected Coursework (Economics): Econometrics, Public Finance, Finance, Intermediate Macroeconomics, Development Economics, Economic History, Game Theory
- Selected Coursework (Quantitative Methods): Statistics, Calculus, Linear Algebra and Probability, Python Programming, Machine Learning and Causal Inference

RESEARCH EXPERIENCE

Can Digital Tax Infrastructure Improve Firms' Loan Contract Design? Evidence from China's Digital Invoice Reform

Independent Research Project, March 2026 – September 2026

- Constructed and harmonized a panel of 20,679 firms and 268,931 firm-year observations across A-share, BSE, and NEEQ markets by linking iFinD, CSMAR, policy-timing, industry, and firm-history data; Retained delisted, transferred, and removed firms to limit sample-selection bias.
- Exploited the province-staggered rollout of fully digital invoices and pre-period industry exposure in a two-tier difference-in-differences design; Validated identification using event studies, modern staggered-DID estimators, placebo tests, leave-one-province-out checks, and wild-cluster-bootstrap inference in Python and R.
- Documented a substantially stronger long-term-debt response among less transparent NEEQ firms, consistent with lower verification costs; assessed competing mechanisms and sensitivity to sample construction.

Research on the Impact of Digital Economy Development on Corporate Debt Maturity Structure and Its Mechanisms: Evidence from Chinese A-share Listed Firms

Undergraduate Researcher, Master's Thesis, Advisor: Prof. Ying Wang, RUC, October 2025 – June 2026

- Matched firm financials to a digital-economy index covering more than 280 Chinese cities, constructing a panel of 19,174 firm-year observations from 2011–2023.
- Utilized DID in Python to estimate two-way fixed-effects, IV-2SLS, and system-GMM models with city-clustered inference and extensive robustness and heterogeneity analyses; Found larger long-term-debt share where digital development is higher, with weaker effects under tighter financing constraints.

Budget Disclosure Practices and Transparency in Chinese County Governments

Research Assistant, Supervisor: Prof. Cong Qin, RUC, September 2025 – January 2026

- Collected and structured budget and final-account disclosures for 100+ Chinese counties, recording disclosure timing, document type, accessibility, and webpage characteristics for research on local fiscal transparency.
- Cleaned and validated the resulting dataset, resolving inconsistent document names, missing disclosure years, broken links, and differences in local reporting formats (Stata).

Tracing the Historical Roots of China's Population Modernization in the Ming and Qing Dynasties

Research Assistant, Supervisor: Prof. Sijie Hu, University of Hong Kong, March 2025 – October 2025

- Constructed and validated a large-scale historical demographic database from Ming-Qing genealogical manuscripts for quantitative analysis of long-run population change.

- Extracted, cleaned, and standardized thousands of individual and kinship records (Excel), including births, marriages, burials, parent-child relationships, spouses, and adoptions, using cross-text verification.
- Developed systematic consistency checks to identify name ambiguity, contradictory family links, duplicate relationships, and missing observations, documenting recurring error patterns for data cleaning.

Pricing and Hedging of Financial Derivatives under Stochastic Volatility Models
Researcher, Supervisor: Dr. Jaksa Cvitanić, California Institute of Technology, January 2025 – May 2025
- Conducted empirical analysis of derivative pricing and hedging using real-market equity and options data under Black-Scholes and stochastic-volatility frameworks.
- Developed Python workflows to scrape, clean, align, and preprocess S&P 500 equity and options datasets for volatility estimation and model calibration.
- Estimated historical and implied volatility and applied Bayesian parameter optimization to stochastic-volatility models, improving fit to market data by more than 15%.

The Impact of China's 12-Year Compulsory Education Reform on Labor Outcomes
Researcher, Supervisor: Prof. Chen Weize, RUC, October 2024 – December 2024
- Investigated how China's 12-year compulsory education reform relates to labor productivity and wage outcomes using econometric analysis in Stata.
- Merged CFPS microdata with regional statistical yearbooks to construct measures of schooling attainment, wages, labor productivity, and local policy exposure in Zhuhai and Zhongshan.
- Conducted heteroskedasticity tests, robust regressions, and LM diagnostics to assess model validity and document relationships among educational attainment, wages, and productivity.

PROFESSIONAL EXPERIENCE

Guotai Haitong Securities | Shanghai, China
Quantitative Analyst Intern, May 2025 – July 2025
- Processed ten years of A-share daily data using Python and constructed factor pools for growth, value, momentum, volatility, and size within a Barra multi-factor framework.
- Developed back-testing modules for cross-sectional grouping, parameter tuning, and long-short portfolios; Evaluated information coefficients, sensitivity, stability, and excess returns.

Baidu Inc. | Beijing, China
Data Analyst Intern, July 2025 – September 2025
- Evaluated DeepSeek and GPT-based large language models across multiple datasets, comparing preprocessing, model design, performance, prompt behavior, and computational efficiency.
- Proposed weighted-sampling and data-augmentation methods to reduce imbalance across high- and low-frequency domains.

SELECT ACTIVITIES & HONORS

Member, Qualitative Methods Workshop, UChicago, 2026-Present
2nd Prize, 4th "Jingcai Dachuang" Beijing College Students Innovation and Entrepreneurship Competition, 2025
3rd Prize, 26th "Innovation Cup" Extracurricular Academic and Technological Works Competition, RUC, 2024